12 LIGHT WAVES

We see objects because they emit or reflect light. In this chapter you will learn how light behaves when it reflects from different surfaces, and what happens when light travels from one transparent material to another.



▲ Figure 12.1 In the Hall of Mirrors at the fairground the reflection of light can be very confusing!

LEARNING OBJECTIVES

- Know that light waves are transverse waves and that they can be reflected and refracted
- Use the law of reflection (the angle of incidence equals the angle of reflection)
- Draw ray diagrams to illustrate reflection and refraction
- Practical: investigate the refraction of light, using rectangular blocks, semi-circular blocks and triangular prisms
- Know and use the relationship between refractive index, angle of incidence and angle of refraction:

$$n = \frac{\sin i}{\sin r}$$

- Practical: investigate the refractive index of glass, using a glass block
- Describe the role of total internal reflection in transmitting information along optical fibres and in prisms
- Explain the meaning of critical angle c
- Know and use the relationship between critical angle and refractive index:

$$\sin c = \frac{1}{n}$$

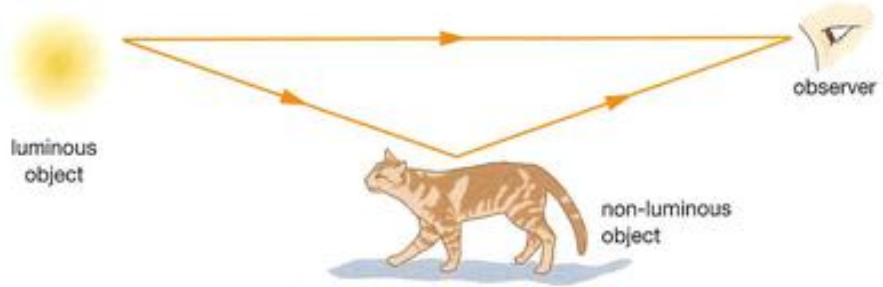


▲ Figure 12.2 Cataracts mean that light cannot enter the eye correctly.

SEEING THE LIGHT

The patient shown in Figure 12.2 has a cataract. The front of one of his eyes has become so cloudy that he is unable to see. Nowadays it is possible to remove this damaged part of the eye and replace it with a clear plastic that will allow light to enter the eye again.

There are many sources of light, including the Sun, the stars, fires, light bulbs and so on. Objects such as these that emit their own light are called luminous objects. When the emitted light enters our eyes we see the object. Most objects, however, are non-luminous. They do not emit light. We see these non-luminous objects because of the light they reflect.



▲ Figure 12.3 Luminous objects, such as the Sun, give out light. Non-luminous objects only reflect light.

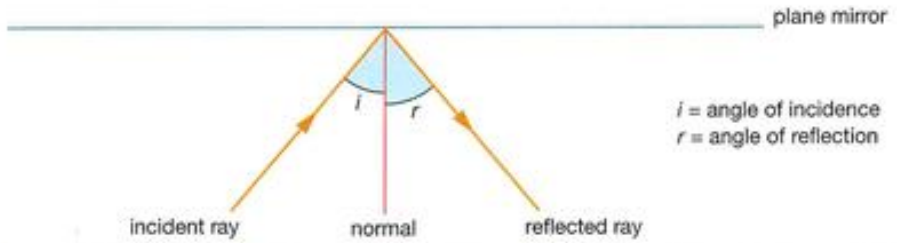
REFLECTION

When a ray of light strikes a plane (flat) mirror, it is reflected so that the angle of incidence (i) is equal to the angle of reflection (r).

KEY POINT

The angle of incidence is the angle between the incident ray and the normal.

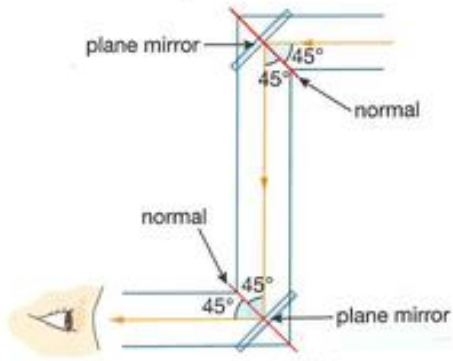
The angle of reflection is the angle between the reflected ray and the normal.



▲ Figure 12.4 Light is reflected from a plane mirror. The angle of incidence is equal to the angle of reflection. The normal is a line at right angles to the mirror.

Mirrors are often used to change the direction of a ray of light. One example of this is the simple periscope, which uses two mirrors to change the direction of rays of light.

Rays from the object strike the first mirror at an angle of 45° to the normal. The rays are reflected at 45° to the normal and so are turned through an angle of 90° by the mirror. At the second mirror the rays are again turned through 90°. Changing the direction of rays of light in this way allows an observer to use a periscope to see over or around objects.



▲ Figure 12.5 A periscope is used to see over or around objects.

REFRACTION



▲ Figure 12.6 This rainbow is caused by refraction.

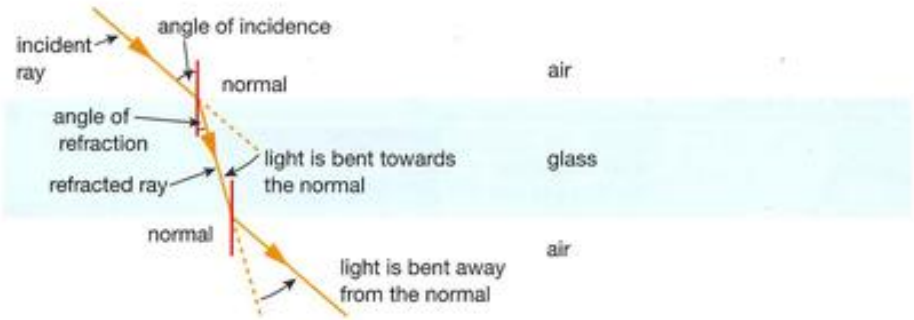
KEY POINT

A medium is a material, such as glass or water, through which light can travel. The plural of medium is media.

KEY POINT

Light does travel more slowly in air than in a vacuum but the difference is tiny.

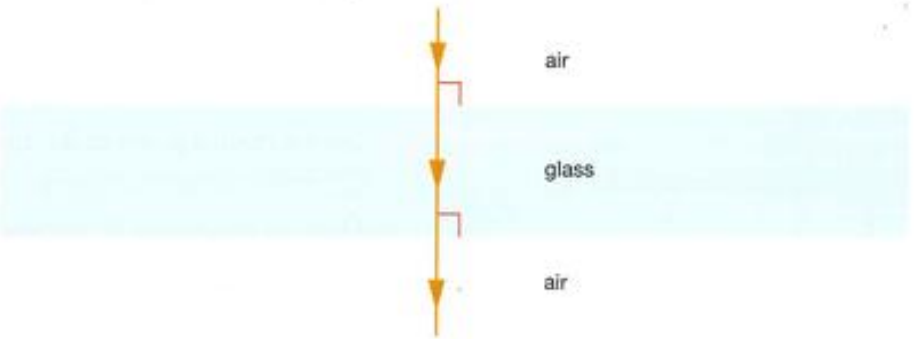
Rays of light can travel through many different transparent media, including air, water and glass. Light can also travel through a vacuum. In a vacuum and in air, light travels at a speed of 300 000 000 m/s. In other media it travels more slowly. For example, the speed of light in glass is approximately 200 000 000 m/s. When a ray of light travels from air into glass or water it slows down as it crosses the border between the two media. This change in speed may cause the ray to change direction. This change in direction of a ray is called refraction.



▲ Figure 12.7 This light ray is being refracted twice – once as it travels from air into glass and then as it travels from glass to air.

As a ray enters a glass block, it slows down and is refracted towards the normal. As the ray leaves the block it speeds up and is refracted away from the normal.

If the ray strikes the boundary between the two media at 90°, the ray continues without change of direction (Figure 12.8).



▲ Figure 12.8 If the light hits the boundary at 90° the ray does not bend.

REFRACTIVE INDEX

Different materials can bend rays of light by different amounts. We describe this by using a number called the **refractive index** (n). The refractive index of glass is about 1.5 and water is 1.3. This tells us that under similar circumstances glass will refract light more than water.

We can use the equation below to calculate the refractive index of a material:

$$n = \frac{\sin i}{\sin r}$$

where i is the angle of incidence and r is the **angle of refraction**.

EXAMPLE 1

In an experiment similar to the one shown in Figure 12.9, the angle of incidence was measured as 30° and the angle of refraction as 19°. Calculate the refractive index of the glass block.

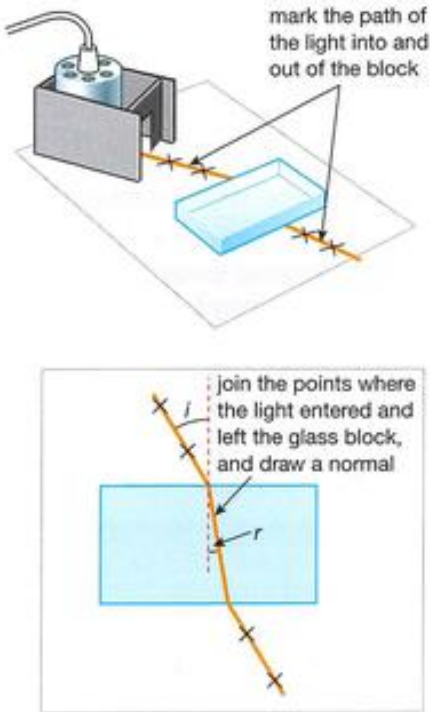
$$n = \frac{\sin i}{\sin r}$$

$$= \frac{\sin 30}{\sin 19}$$

$$= \frac{0.5}{0.33}$$

$$= 1.52$$

! Ray box lamps get hot enough to burn skin and char paper. Glass blocks and prisms should be handled carefully and not knocked together – they can splinter or shatter.



▲ Figure 12.9 How to investigate refraction using a rectangular glass block

ACTIVITY 1

▼ PRACTICAL: INVESTIGATE THE REFRACTIVE INDEX FOR GLASS

You can investigate the refractive index of glass using a ray box and a rectangular glass block.

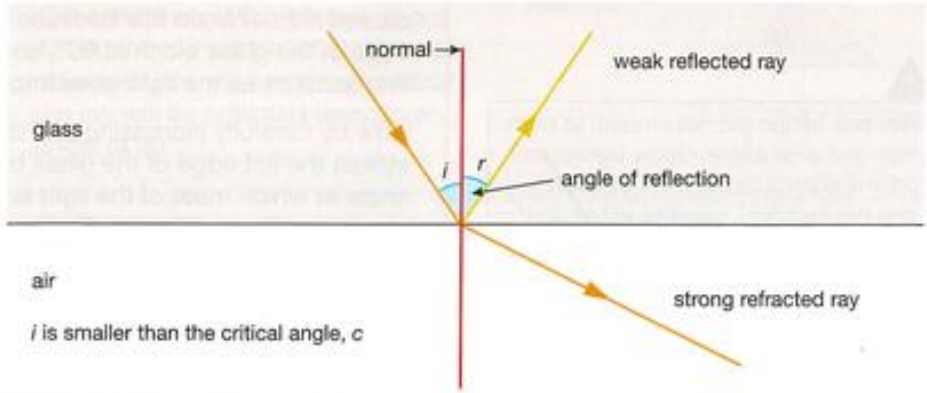
1. Shine a ray of light onto one of the sides of the glass block, so that the ray emerges on the opposite side of the block. Mark the directions of both of these rays with crosses.
2. Draw around the glass block before removing it.
3. Using the crosses, draw in the direction of both rays.
4. Draw in the direction of the ray that travelled inside the glass block.
5. Draw a normal (a line at 90° to the glass surface) where the ray enters the block.
6. Measure the angles of incidence (i) and refraction (r) (see Figure 12.9).
7. Use the equation $n = \frac{\sin i}{\sin r}$ to find the refractive index of the glass block.

TOTAL INTERNAL REFLECTION

When a ray of light with a small angle of incidence passes from glass into air, most of the light is refracted away from the normal but if we look carefully we can see that there is a small amount that is reflected from the boundary. Total internal reflection only occurs when rays of light are travelling towards a boundary with a less optically dense medium (a medium with a lower refractive index).

KEY POINT

The symbol '*r*' is used for both the angle of reflection and the angle of refraction.

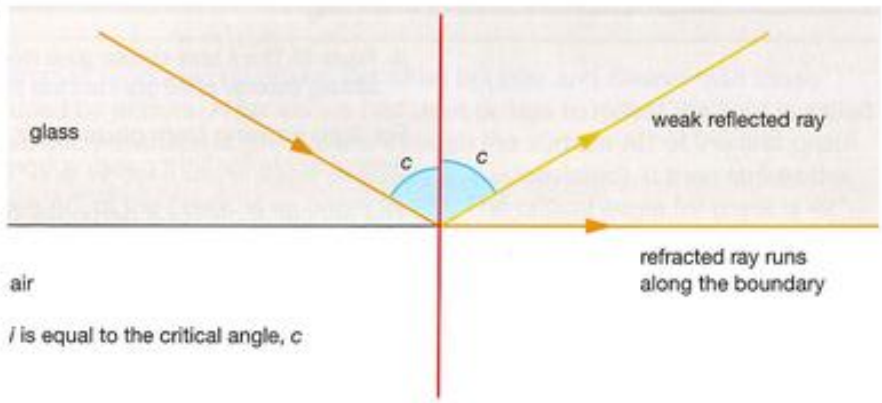


▲ Figure 12.10 A ray of light travelling from glass to air

But as the angle of incidence in the glass increases, the angle of refraction also increases until it reaches a special angle called the critical angle (*c*). The angle of refraction now is 90°.

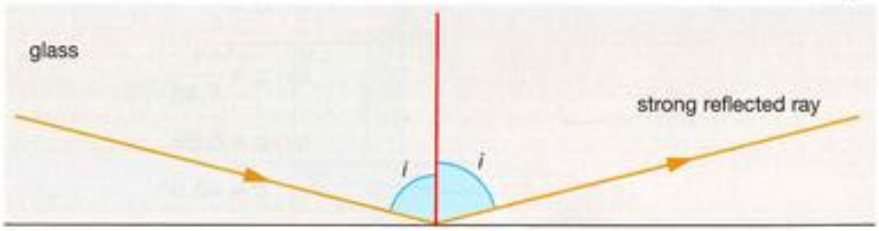
The critical angle is the smallest possible angle of incidence at which light rays are totally internally reflected.

a



▲ Figure 12.11a Ray of light strikes glass/air boundary at the critical angle

b



▲ Figure 12.11b When *i* is greater than *c* total internal reflection occurs.

When *i* is greater than the critical angle, all the light is reflected at the boundary. No light is refracted. The light is totally internally reflected.

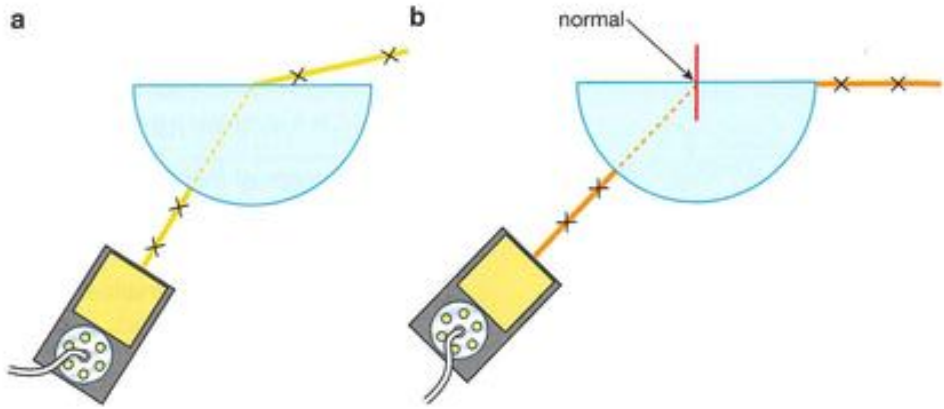
ACTIVITY 2

▼ PRACTICAL: INVESTIGATE TOTAL INTERNAL REFLECTION

You can investigate total internal reflection in the laboratory using a semi-circular glass block and a ray box. As shown in Figure 12.12a, a ray of light is directed at the centre of the straight side of the block through the curved side. (We do this because the incident ray will then always hit the edge of the glass block at 90°, so there are no refraction effects to take into account as the light goes into the block.)

Now by carefully increasing and decreasing the angle at which the ray strikes the flat edge of the glass block, we can discover the smallest angle at which most of the light is refracted along the edge of the glass block (see Figure 12.12b). This angle is the critical angle.

! Ray box lamps get hot enough to burn skin and char paper. Glass blocks and prisms should be handled carefully and not knocked together – they can splinter or shatter.



▲ Figure 12.12 a A semi-circular glass block used to demonstrate total internal reflection b Light striking the edge of the glass block at the critical angle

For light passing from glass to air, the critical angle is typically 42° and the critical angle for light passing from water to air is 49°.

The critical angle for a particular medium is related to its refractive index by this equation:

$$\sin c = \frac{1}{n}$$

EXAMPLE 2

The refractive index for a type of glass is 1.45. Calculate the critical angle.

$$\sin c = \frac{1}{n}$$

$$\sin c = \frac{1}{1.45}$$

$$\sin c = 0.69$$

$$c = 43.6°$$

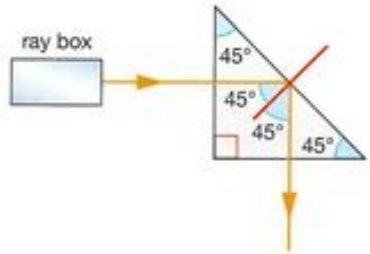
We sometimes use **prisms** rather than mirrors to reflect light. The light is totally internally reflected by the prism.

ACTIVITY 3

▼ PRACTICAL: INVESTIGATE TOTAL INTERNAL REFLECTION IN PRISMS

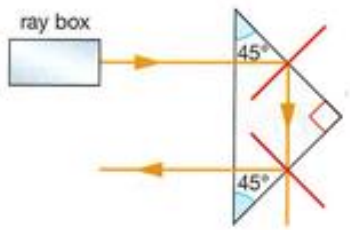
> ! Ray box lamps get hot enough to burn skin and char paper. Glass blocks and prisms should be handled carefully and not knocked together – they can splinter or shatter.

If you shine a ray of light into a prism as shown in Figure 12.13 it will strike the far surface at an angle of 45°. The critical angle for glass is about 42° so the ray will be totally internally reflected. You will see therefore that the ray will be reflected through an angle of 90°.



▲ Figure 12.13 Turning through 90° using total internal reflection

If you shine a ray into the prism as shown in Figure 12.14 the ray will be reflected through an angle of 180° – that is, it will go back in the direction from which it came.

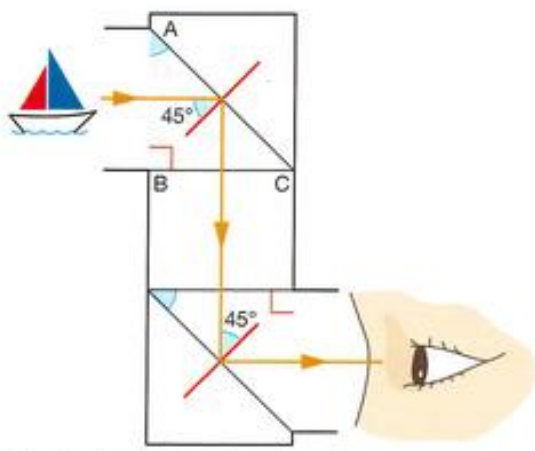


▲ Figure 12.14 Turning through 180° using total internal reflection

USING TOTAL INTERNAL REFLECTION

THE PRISMATIC PERISCOPE

The images produced by prisms are often brighter and clearer than those produced by mirrors. A periscope that uses prisms to reflect the light is called a prismatic periscope. Light passes through the surface AB of the first prism at 90° and so does not change direction (it is undeviated). It then strikes the surface AC of the prism at an angle of 45°. The critical angle for glass is 42° so the ray is totally internally reflected and is turned through 90°. When it leaves the first prism the light travels to a second prism. The second prism is positioned so that the ray is again totally internally reflected. The ray emerges parallel to the direction in which it was originally travelling.



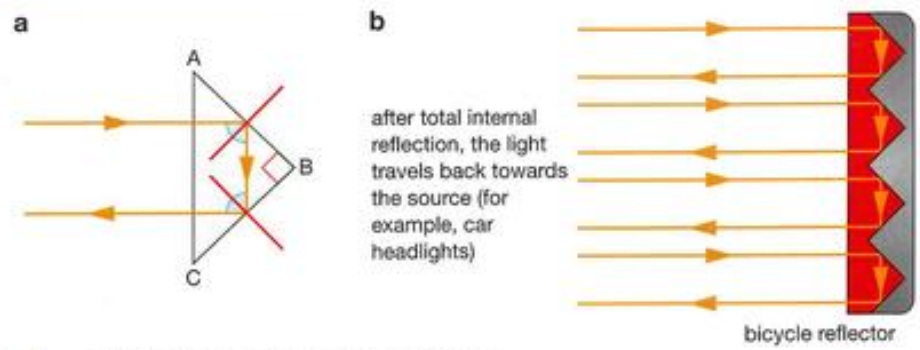
▲ Figure 12.15 Total internal reflection in a prismatic periscope

BICYCLE AND CAR REFLECTORS



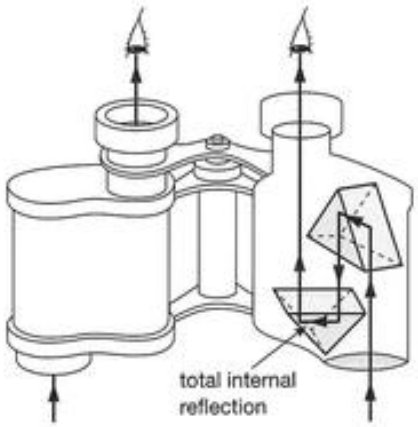


▲ Figure 12.17 Reflectors like these can save lives.



▲ Figure 12.16 Prisms can also be used as reflectors.

Light entering the prism in Figure 12.16 is totally internally reflected twice. It emerges from the prism travelling back in the direction from which it originally came. This arrangement is used in bicycle or car reflectors.



▲ Figure 12.18 Total internal reflection inside binoculars



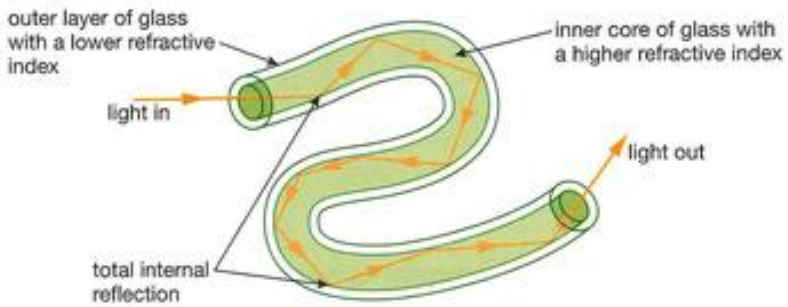
▲ Figure 12.19 Prismatic binoculars

Binoculars also make use of total internal reflection within prisms.

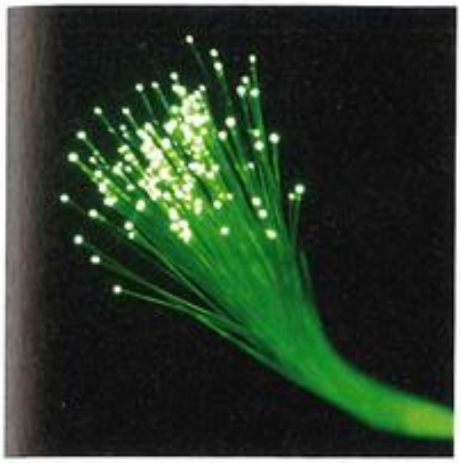
Each side of a pair of binoculars contains two prisms to totally internally reflect the incoming light. Without the prisms, binoculars would have to be very long to obtain large **magnifications** and would look like a pair of telescopes.

OPTICAL FIBRES

One of the most important **applications** for total internal reflection is the optical fibre. This is a very thin piece of fibre composed of two different types of glass. The centre is made of a glass that has a high refractive index surrounded by a different type of glass that has a lower refractive index.



▲ Figure 12.20 In an optical fibre, light undergoes total internal reflection.



▲ Figure 12.21 Optical fibres

As the fibres are very narrow, light entering the inner **core** always strikes the boundary of the two glasses at an angle that is greater than the critical angle. No light escapes across this boundary. The fibre therefore acts as a 'light pipe' providing a path that the light follows even when the fibre is curved.

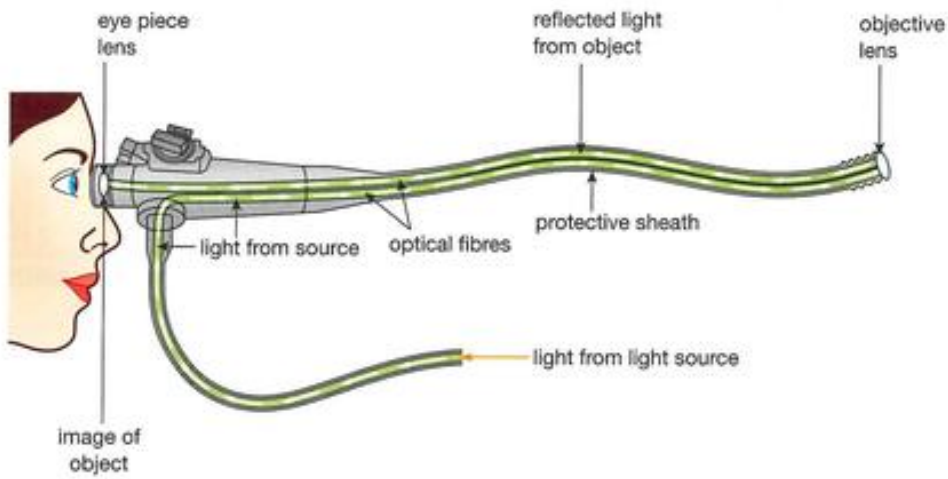
Large numbers of these fibres fixed together form a bundle. Bundles can carry sufficient light for images of objects to be seen through them. If the fibres are tapered (narrower at one end) it is also possible to produce a magnified image.

Figure 12.22 shows optical fibres in an endoscope. The endoscope is used by doctors to see the inside the body – for example, to examine the inside of the stomach. Endoscopes can also be used by engineers to see hard-to-reach parts of machinery.

Light travels down one bundle of fibres and shines on the object to be viewed. Light reflected by the object travels up a second bundle of fibres. An image of the object is created by the eyepiece.

THE ENDOSCOPE

By using optical fibres to see what they are doing, doctors can carry out operations through small holes made in the body, rather than through large cuts. This is called 'keyhole surgery'. This is less stressful for patients and usually leads to a more rapid recovery.



▲ Figure 12.22 Optical fibres are used in endoscopes to see inside the body.

OPTICAL FIBRES IN TELECOMMUNICATIONS

Modern telecommunications systems use optical fibres rather than copper wires to transmit messages as less energy is lost. Electrical signals from a telephone are converted into light energy produced by tiny lasers, which send pulses (small amounts) of light into the ends of optical fibres. A light-sensitive detector at the other end changes the pulses back into electrical signals, which then flow into a telephone receiver (ear piece).

CHAPTER QUESTIONS

More questions on refraction can be found at the end of Unit 3 on page 130.

SKILLS INTERPRETATION, PROBLEM SOLVING (6)

1 Draw a ray diagram to show how a ray of light can be turned through 100° using two plane mirrors. Mark on your diagram a value for the angle of incidence at each of the mirrors.

SKILLS INTERPRETATION

2 a Draw a diagram to show the path of a ray of light travelling from air into a rectangular glass block at an angle of about 45°.

b Show the path of the ray as it emerges from the block.

SKILLS CRITICAL THINKING

c Explain why the ray changes direction each time it crosses the air/glass boundary.

SKILLS INTERPRETATION

d Draw a second diagram showing a ray that travels through the block without its direction changing.

SKILLS PROBLEM SOLVING (7)

3 In an experiment to measure the refractive index of a type of glass, the angle of refraction was found to be 31° when the angle of incidence was 55°.

a Calculate the refractive index of the glass.

b What would the angle of refraction be for a ray with an angle of incidence of 45°?

c Calculate the critical angle for the glass.

SKILLS INTERPRETATION (6)

4 a Draw a diagram to show how a prism can create a rainbow of colours.

SKILLS CRITICAL THINKING

b Explain how these colours are produced by the prism.

SKILLS INTERPRETATION (7)

5 Draw three ray diagrams to show what happens to a ray of light travelling in a glass block in the following situations. It hits a face of the block at an angle:

a less than the critical angle

b equal to the critical angle

c greater than the critical angle.

SKILLS CRITICAL THINKING (6)

6 a What is meant by 'total internal reflection of light' and under what conditions does it occur?

SKILLS INTERPRETATION

b Draw a diagram to show how total internal reflection takes place in a prismatic periscope.

SKILLS REASONING

c Give one advantage of using prisms in a periscope rather than plane mirrors.

SKILLS INTERPRETATION (7)

d Draw a second diagram to show how a prism could be used to turn a ray of light through 180°. Give one application of a prism used in this way.

SKILLS INTERPRETATION, CRITICAL THINKING

7 a Explain why a ray of light entering an optical fibre is unable to escape through the sides of the fibre. Include a ray diagram in your explanation.

SKILLS CRITICAL THINKING

b Explain how doctors use optical fibres to see inside the body.

(6)

c Name one other use of optical fibres.